

B.Tech. (V Sem.)

23AD55- DATA MINING LAB

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Pre-requisites: Data Base Management Systems, Python Programming/R Programming

Course Objectives: The objective of this lab is to provide practical exposure to the implementation of widely used data mining algorithms and to develop the ability to evaluate their performance in both supervised and unsupervised learning environments.

Software Requirements: Python/R-

Tool Course Outcomes

CO1: Apply data preprocessing, similarity measures, and machine learning models using Python for solving data mining tasks.

CO2: Develop and evaluate classification, clustering, and association rule mining techniques using appropriate algorithms.

CO3: Analyze performance of data mining models using validation techniques and interpret results for decision-making.

CO4: Improve individual/teamwork skills, communication, report writing, and practice ethical responsibilities in project work.

List of Experiments:

Use python library scikit-learn wherever necessary

1. Demonstrate the following data preprocessing tasks using python libraries.
 - a) Loading the dataset
 - b) Identifying the dependent and independent variables.
 - c) Dealing with missing data
2. Demonstrate the following data preprocessing tasks using python libraries.
 - a) Dealing with categorical data.
 - b) Scaling the features.
 - c) Splitting dataset into Training and Testing Sets
3. Demonstrate the following Similarity and Dissimilarity Measures using python
 - a) Pearson's Correlation
 - b) Cosine Similarity
 - c) Jaccard Similarity
 - d) Euclidean Distance
 - e) Manhattan Distance
4. Build a model using linear regression algorithm on any dataset.
5. Build a classification model using Decision Tree algorithm on iris dataset
6. Apply Naïve Bayes Classification algorithm on any dataset
7. Generate frequent itemset using Apriori Algorithm in python and also generate association rules for any market basket data.
8. Apply K- Means clustering algorithm on any dataset.
9. Apply Hierarchical Clustering algorithm on any dataset.
10. Apply DBSCAN clustering algorithm on any dataset.

Web Resources:

1. <https://analyticsindiamag.com/data-pre-processing-in-python/>
2. <https://towardsdatascience.com/decision-tree-in-python-b433ae57fb9>
3. <https://towardsdatascience.com/calculate-similarity-the-most-relevant-metrics-in-anutshell-9a43564f533e>

4. <https://www.springboard.com/blog/data-mining-python-tutorial/>
5. <https://medium.com/analytics-vidhya/association-analysis-in-python-2b955d0180c>
6. <https://www.datacamp.com/community/tutorials/naive-bayes-scikit-learn>
7. <https://www.analyticsvidhya.com/blog/2019/05/beginners-guide-hierarchicalclustering/>
8. <https://towardsdatascience.com/dbscan-algorithm-complete-guide-and-application-withpython-scikit-learn690cbae4c5d>